



UNIVERSITY OF THE PUNJAB

Third Semester 2015
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Statistics-III

Course Code: STAT-201

TIME ALLOWED: 2 hrs. & 30 mins.

MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Q.2. Write short answers to the following questions.

(20 Marks) 2 each

- ✓(i) What is the difference between sampling error and standard error?
- ✓(ii) Differentiate between the simple and composite hypotheses?
- ✓(iii) Discuss type-I and type-II error?
- ✓(iv) What is precision?
- ✓(v) Why statistic is a variable and parameter is a constant?
- ✓(vi) Discuss the term non-response?
- ✓(vii) Write down the assumptions of t-distribution.
- ✓(viii) What are the properties of a good point estimator? Discuss any one of them.
- ✓(ix) Define the terms estimation, Estimator and Estimate.
- ✓(x) What is Bias? How it occurs in sample survey.

Q.3. The random variable X has the following probability distribution:

(5)

X	4	5	6	7
P(X)	0.25	0.35	0.30	0.10

Find the probability that the mean of 25 items will be less than 5.5.

Q.4. If $x_1, x_2, \dots, x_n$ be a random sample of size 'n' taken from a Normal population (μ, σ^2) find Maximum Likelihood Estimator of σ^2 when μ is known. (5)

Q.5. (a) A random sample of 500 members of the labor on a certain region showed that 43 were unemployed. Construct 90% Confidence Interval for proportion of unemployed on the region. (2.5)

(b) If X is normally distributed and given the sample values as $\bar{X} = 42, S = 10, n = 25$. Find 98% confidence Interval for σ^2 . (2.5)

Q.6. Two random samples taken independently from normal population with identical variances yield the results as (5)

$$n_1 = 12, \bar{x}_1 = 10, s_1^2 = 1200, n_2 = 20, \bar{x}_2 = 18, s_2^2 = 1000,$$

Test the hypotheses that the difference between population means is 10 against alternative that it is less than 10 at 5% level of significance.

Q.7. Following calculations have been obtained from two samples selected from normally distribution populations. (5)

$$n_1 = 11, n_2 = 6, \bar{x}_1 = 50.3, \bar{x}_2 = 37.8, S_1^2 = 46.74, S_2^2 = 51.47,$$

Test whether the two populations have the same variances at 5% level of significance.



UNIVERSITY OF THE PUNJAB

Third Semester 2017
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Statistics-III
Course Code: STAT-201/STT-21314

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Part-2 (SUBJECTIVE)

Note: Attempt ALL questions.

(2x10)

Q.3. Write short answers to each of the followings.

- ✓ i) What is the object of sampling? (Notes) pg (3)
- ✓ ii) Differentiate between target and sampled population. Notes (1)
- ✓ iii) Differentiate between random and simple random sampling. II pg (29) (30) (1.5)
- ✓ iv) In what sense unbiasedness differs from consistency? Notes
- v) What is meant by mean square error of an estimator?
- vi) What is the purpose of goodness-of-fit test?
- vii) Why is the z-test usually inappropriate as a test statistic when sample size is small?
- viii) Under what conditions, the paired t-test is used?
- ✓ ix) What do you mean by level of significance?
- ✓ x) What do you mean by p-value?

Q.4. Suppose a population of $N = 9$ is stratified into 3 strata with the following measurements:

(06) pg (19)

Stratum I	$X_{11} = 1, X_{12} = 2, X_{13} = 4$
Stratum II	$X_{21} = 5, X_{22} = 8$
Stratum III	$X_{31} = 11, X_{32} = 15, X_{33} = 16, X_{34} = 19$

If two measurements are drawn from each stratum for the sample, state how many samples of size 6 could be chosen from this population? List these samples and compute the mean for each sample.

Q.5. The following table shows the distribution of 14-year-old schoolboy intelligence test markings.

(06)

IO	80-89	90-99	100-109	110-119	120-129	130-139	140-149
Number	30	52	75	109	65	42	27

On the assumption that this group is a random sample, estimate the standard error of the mean and explain its usefulness.

Q.6. Let the simple linear regression be $Y_i = \hat{\alpha} + \hat{\beta}X_i + e_i$, where each Y_i is normally distributed and X_i 's are fixed. Then show (prove) that $\hat{\alpha}$ and $\hat{\beta}$ are unbiased estimators of the parameters α and β .

(06)

Q.7. A form of intelligence test was given to random samples of soldiers and sailors in a certain country. The following results were recorded:

(06)

	Number in Samples	Mean Score	Sample Standard Deviation
Soldiers	332	12.78	2.43
Sailors	615	12.99	2.48

Assume the populations of scores to be normal. What conclusion should be drawn?

Q.8. To verify whether a course in statistics improved performance, a similar test was given to 12 participants both before and after the course. The original grades recorded in alphabetical order of the participants were 44, 40, 61, 52, 32, 44, 70, 41, 67, 72, 53 and 72. After the course, the grades were in the same order 53, 38, 69, 57, 46, 39, 73, 48, 73, 74, 60 and 78.

(06)

Was the course useful, as measured by performance on the test? Consider these 12 participants as a sample from a population.

18.28



UNIVERSITY OF THE PUNJAB

Third Semester 2018
Examination: B.S. 4 Years Programme

Roll No.

PAPER: Statistics-III
Course Code: STAT-201/STT-21314

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Part-2 (SUBJECTIVE)

Note: Attempt ALL questions.

- Q.3. Write short answers to each of the followings. Marks
- i) Sampling is superior than complete enumeration, discuss. (04) saves money wala.
 - ii) In population census cluster sampling is preferred over stratified random sampling. why? (02)
 - iii) Compare random and simple random sampling. (02)
 - iv) Which is more dangerous, sampling error or non-sampling error? Why? (03)
 - v) What is meant by mean square error of an estimator? (02)
 - vi) What is the purpose of goodness-of-fit test? (02)
 - vii) Why is the z-test usually inappropriate as a test statistic when sample size is small? (03)
 - viii) What do you mean by p-value? (02)
- Q.4. Suppose a population of $N = 9$ is stratified into 3 strata with the following measurements. (06)

Stratum I	$X_{11} = 1, X_{12} = 2, X_{13} = 4$
Stratum II	$X_{21} = 6, X_{22} = 8$
Stratum III	$X_{31} = 11, X_{32} = 15, X_{33} = 16, X_{34} = 19$

If two measurements are drawn from each stratum for the sample, state how many samples of size 6 could be chosen from this population? List these samples and compute the mean for each sample.

- Q.5. Let X_1, X_2, X_3 , and X_4 be a random sample of size $n = 4$ from a $N(\mu, \sigma^2)$. A statistician wishes to estimate the mean by using either of the following two estimators of the mean μ . (06)

$$T_1 = \frac{X_1 + X_2 + X_3 + X_4}{4} \quad (\text{the sample mean, } \bar{X})$$

$$J_2 = \frac{X_1 + 2X_2 + 3X_3 + X_4}{7} \quad (\text{a weighted mean, } \bar{X}_w)$$

Which estimator should be preferred?

- Q.6. A sample of readings from a normal population with unknown mean μ and unknown variance σ^2 gave the following data: (06)

x	17.4	17.5	17.6	17.7	17.8
f	12	16	19	23	10

A second sample of readings taken from the same population gave $n_2 = 72$, $\sum X = 1267.2$, $\sum X^2 = 22536$. Combine the two samples to give estimates μ and σ^2 , and give the approximate 90% confidence interval for μ . P.T.O.

@ In cluster sampling, the aim is to reduce cost and increase the efficiency of sampling. In stratified sampling, the motive is to increase precision to reduce error.

Q.7. A random sample of size 40 from non-normal population yielded the $\bar{X}_1 = 70.4$, $S_1^2 = 31.40$ another random sample of size 50 from second non normal population yielded $\bar{X}_2 = 65.3$, $S_2^2 = 44.82$, test the hypothesis that the difference between two means is equal to 2. (06)

Q.8. Ten young recruits were put through a strenuous physical training program by the Army. Their weights were recorded before and after the training with the following results: (06)

Recruit	1	2	3	4	5	6	7	8	9	10
Weight before	125	195	160	171	140	201	170	176	195	139
Weight after	136	201	158	184	145	195	175	190	190	145

Using $\alpha = 0.05$, would you say that the program affects the average weight of recruits? Assume the distribution of weights before and after to be approximately normal.



UNIVERSITY OF THE PUNJAB

Third Semester – 2019

Examination: B.S. 4 Years Program

Roll No.

PAPER: Statistics-III

Course Code: STAT-201/STT-21314 Part – II

MAX. TIME: 2 Hrs. 45 Min.

MAX. MARKS: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

- Q.2) Write short answers to each of the followings. (Marks)
- ✓ i) What are the advantages and disadvantages of using this sampling technique? (04)
 - ✓ ii) Differentiate between probability sampling and non-probability sampling. What are their relative advantages and disadvantages? (04)
 - ✓ iii) Differentiate between Multistage and Multi phase Sampling. (03)
 - ✓ iv) What is a sampling distribution? Describe the properties of the sampling distribution of the means. (05)
 - ✓ v) Under what conditions, the paired t-test is used? (02)
 - ✓ vi) What do you mean by p-value? (02)

- Q.3. Using a random number table, select 30 samples of size 3 each with replacement from the following population distribution of heights. Find the mean of sample means. (06)

Height (inches)	No. of students
60 - 62	5
63 - 65	18
66 - 68	42
69 - 71	27
72 - 74	8

- Q.4. A sample of 40 observations from a population with unknown mean μ and unknown variance σ^2 gave $\sum X = 800$ and $\sum X^2 = 16052$. A second sample taken from the same population gave

x	15	19	20	21	22	23
f	12	23	10	7	5	3

Using the data given by the two samples, find unbiased estimates of μ and σ^2 .

- Q.5. A firm wishes to test the hypothesis that the average monthly wage of its shop employees is Rs 180 with a standard deviation of Rs.4.50. They wish to run a risk of only 0.05 or rejecting the null hypothesis when it is true, and a risk of only 0.05 of accepting the null hypothesis if the average is as high as Rs. 182 or as low as Rs 178. How large a sample should be taken? (06)
- Q.6. A random sample of size 40 from non-normal population yielded the $\bar{X}_1 = 70.4$, $S_1^2 = 31.40$ another random sample of size 50 from second non normal population yielded $\bar{X}_2 = 65.3$, $S_2^2 = 44.82$, test the hypothesis that the difference between two means is equal to 2. (06)
- Q.7. A random sample of 250 men and 250 women were polled as to their desire concerning the ownership of television sets. The following data resulted. (06)

Classification	Men	Women	Total
Want television	80	120	200
Don't want television	170	130	300
Total	250	250	500

Test the hypothesis that desire to own a television set is independent of sex at the 0.01 level of significance.



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Third Semester – Fall 2022

Paper: Statistics-III

Course Code: STAT-201

Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(15x2=30)

- i. What are the advantages of sampling?
- ii. Differentiate between random sampling and simple random sampling.
- iii. Differentiate between Parameter and statistic.
- iv. What is the difference between sampling with replacement and sampling without replacement?
- v. What is meant by the term "Unbiasedness"?
- vi. Is an estimator a random variable? Why or why not?
- vii. Differentiate between simple Hypothesis and composite Hypothesis.
- viii. What do you know about the term level of significance?
- ix. Differentiate between One tailed test and Two tailed test
- x. What do you know about Rejection Region?
- xi. What are the assumptions in using the t- distribution?
- xii. Write down any 2 properties of χ^2 distribution.
- xiii. Define the term STRATUM.
- xiv. What do you know about Mean Square Error of an estimator?
- xv. What is meant by P-values?

Q.2. Answer the following questions.

(5x6=30)

- i. At a small private college, the students are classified as under:

<u>Classification:</u>	B.Sc.,	B.A.,	F.Sc.,	F.A
<u>No of Students:</u>	150,	163,	195,	220

If we wish to select a stratified random sample of size $n = 40$ by proportional allocation, how large a sample must we take from each stratum?

- ii. A random sample of 75 college students is selected and 16 are found to have cars on campus. Find 90% confidence interval for the proportion of students who have cars on campus.
- iii. Can you reject a claim that the average age of members of parliament is at least 50 years? If a random sample of 36 members has a mean age of 48.7 years and a standard deviation of 3.1 years.
- iv. Give the following data
 $n_1 = 12, \bar{x}_1 = 10, s_1^2 = 1200$
 $n_2 = 18, \bar{x}_2 = 25, s_2^2 = 900$
Test the hypothesis that $\mu_2 - \mu_1 > 10$ at $\alpha = 0.05$
- v. A random sample of size 10 has the values as : 67.50, 70.75, 72.00, 63.25, 65.25, 68.75, 69.25, 68.50, 66.50 and 64.75
Test the hypothesis that $\sigma^2 > 7.7953$ at $\alpha = 0.01$

UNIVERSITY OF THE PUNJAB

S. 4 Years Program : Third Semester – Fall 2021

Statistics-III

Course Code: STAT-201

Roll No.

Time: 3 Hrs. Marks: 60

Answer the following short questions:

(15x2=30)

- i. ✓ Differentiate between the terms population and sample. *notes*
- ii. ✓ Define the term BIAS. *notes*
- iii. ✓ Define the term STRATUM.
- iv. ✓ Differentiate between Random Sampling and simple random sampling.
- v. ✓ What is meant by proportional allocation?
- vi. ✓ Is an estimator a random variable? Why or why not?
- vii. ✓ Differentiate between point estimate and interval estimate.
- viii. ✓ What do you know about consistent estimator?
- ix. ✓ Define the term Alternative Hypothesis.
- x. ✓ What is meant by Two Tailed Test? *114 pg*
- xi. ✓ What do you know about Level of confidence?
- xii. ✓ What is meant by Type II error?
- xiii. ✓ What do you know about Rejection Region?
- xiv. ✓ What is meant by the term Statistical inference?
- xv. Write down any 2 properties of the F-distribution?

Answer the following questions.

(5x6=30)

A population consisting of 1000 men has a height distribution with $\sigma = 3$. Find the standard error of mean height for a random sample of 50 men selected

- a) With replacement b) without replacement

If x_1, x_2, x_3 is a random sample from a normal population with mean μ and variance σ^2 and $T_1 = \frac{x_1 + 2x_2 + x_3}{4}$ and $T_2 = \bar{x}$ are the two estimators for μ , are these two estimators unbiased?

A random sample of 25 hens from a normal population showed that the average laying is 250 eggs per year with a variance of 625 eggs. The company claimed that the average laying is at least 235 eggs per year. Test the claim of the company at $\alpha = 0.05$.

The following data gives paired yields of two varieties of wheat. Each pair was planted in a different locality. Test the hypothesis that the mean yields are equal at $\alpha = 0.05$.

Variety I:	45, 32, 58, 57, 60, 38, 47, 51, 42, 38
Variety II:	47, 34, 60, 59, 63, 44, 49, 53, 46, 41

$n_1 = 10$
 $n_2 = 10$

A random sample of 30 values has a variance of 108.55. Test the hypothesis that the population variance (σ^2) is greater than 105 at $\alpha = 0.05$.

(i) STRATUM

When a heterogeneous population is divided into homogeneous subgroups then these subgroups are called strata. Stratum is singular of strata.

⑥ An estimator is a special case of a statistic, a number computed from a sample. Because the value of the estimator depends on the sample, the estimator is a random variable, and the estimate typically will not be equal to the value of the population parameter.

⑦ An estimator of a given parameter is said to be consistent if it converges in probability to the true value of the parameter as the sample size tends to infinity.